

TERMINAL — Functional Specification Document v2.1

The digital infrastructure for hiring heavy assets in Africa v2.1 · June 2026 · Author: Nwabueze Chigozirim Victor (Founder) with Claude Status: ✓ Reference edition for development Companion docs: 01 Product Definition · 02 System Lexicon · 03 MVP Scope · 04 Architecture · 05 Asset Spec Schemas · 07 TSD · 08 Design System (+ design-system/ chapters) · ux/ (journeys, flows, screens) · DECISIONS.md

1. Document Control

Field	Detail
Title	Terminal — Functional Specification Document
Version	2.1 (expanded reference edition; supersedes v2.0 draft and FSD v1.0 May 2026)
Scope	Complete MVP functional behaviour: Hirer mobile app ("Terminal app"), Supplier Portal, Backend API, Ops Console
Audience	The founder; coding agents implementing the platform; QA validating behaviour
Authority	This FSD + DECISIONS.md are authoritative for <i>what the system does</i> . The TSD (doc 07) is authoritative for <i>how</i> . Where this document conflicts with Policy Bible v1.0 or FSD v1.0, this document wins .
Vocabulary	Normative per 02_System_Lexicon; quick reference in §2.2 below

1.1 How to read this document (for coding agents)

- **MUST** / **never** / **always** statements are requirements. Tables of states, rates, and rules are normative.
- Each module section ends with **Acceptance checks** — testable assertions the implementation must satisfy.
- Decision IDs (D-001...D-016) trace to DECISIONS.md; do not re-litigate them in code review.

2. Product Summary

Terminal is a map-first B2B marketplace where verified **Suppliers** list assets across five classes — Plant & Machinery, Trucks & Haulage, Warehousing & Storage, Terminals & Container Yards, Land & Staging — and **Hirers** discover, compare, and pay for hires through the platform. Terminal owns no assets, arranges no logistics, and is not a party to the hire agreement; its product is the **transaction record**: discoverable supply, verified counterparties, locked commercial terms, paid through Terminal, with a documented handover trail.

MVP core loop (everything serves this):

Supplier lists assets at a Yard → Hirer discovers on the Map → Hire requested → Supplier accepts → Hirer pays via Paystack → On Hire → Off-Hire → Completed, payout queued.

Four product truths that shape every requirement: supply is fleets (Supplier → Yard → Listing → Unit is native); trust is the product (verification, locked terms, event logs are first-class); leakage is countered by value, not blocking (payment protection + contact masking until paid); liquidity is corridor-shaped (dual Lagos corridors, D-007).

2.1 What Terminal is NOT

Not an asset owner · not a logistics company · not a bank (collect-only payments; licensed PSP rails from Phase 2) · not a sales platform (hire/lease only) · not an employer of operators or drivers (PB §12 liability positions)

carried forward).

2.2 Vocabulary quick reference (normative: doc 02)

Term	Meaning
Supplier / Hirer	The two market sides (<code>is_supplier</code> / <code>is_hirer</code> ; dual = both flags)
Hire	The transaction (v1 "booking"). Lifecycle states in §7.3
Listing	The published offer of an asset; Asset = the physical thing; Unit = one machine within a multi-unit listing
Yard	A supplier's named location holding listings. Site is reserved for the hirer's project location
Live	Published listing status (never "active" — that word belongs to nothing; the hire in-use state is On Hire)
Service Fee	Terminal's revenue (user-facing term; "commission" internal only). Supplier-pays (D-005), supplier-confidential from hirers (D-014)
Hire Value / Supplier Payout	What hirer pays / what supplier receives
Storefront	A supplier's public company page
Ops Console	The founder's Django-Admin operations surface
Conversation / Enquiry	Messaging objects (enquiry = pre-hire, listing- or storefront-level)
Handover Record	Documented on-hire / off-hire evidence object
Verification	User-facing word for KYC; levels Basic → Verified → Business Verified

3. Business Rules — Money

3.1 Service Fee (D-002, D-008, D-014)

Rates by asset class and winning duration scheme:

Asset class	Daily	Weekly	Monthly
Plant & Machinery	12%	10%	8%
Trucks & Haulage	11%	9%	7%
Warehousing & Storage	10%	8%	6%
Terminals & Container Yards	10%	8%	6%
Land & Staging	10%	8%	6%

Rules:

- Best-price rule (D-008):** `hire_value = min(daily×d, weekly×ceil(d/7), monthly×ceil(d/30))` computed over only the rate schemes the supplier has set. The scheme producing the minimum sets the fee tier. Ties resolve to the longer scheme (lower rate). `d = end_date - start_date + 1` (inclusive).
- Fee floor:** `service_fee = max(tier_rate × hire_value, ₦2,500)`. No flat-fee override; no minimum hire value (D-004).
- Supplier pays the fee (D-005):** hirer pays `hire_value` exactly; `payout_amount = hire_value - service_fee`.
- Supplier-confidential (D-014):** hirers see only the total they pay. No service-fee or payout figure may render on any hirer surface (request preview, hire detail, receipts, emails). The full breakdown (Hire Value / Service Fee / You Receive) renders on supplier and Ops surfaces only. ToS discloses the fee model in

legal terms.

5. **Locking:** `hire_value`, `service_fee`, `payout_amount`, `fee_basis` (e.g. "10% weekly (min ₦2,500)") are computed at request time for preview, **locked at acceptance, immutable thereafter**. Rate-table changes never affect existing hires.
6. Fee applies to hire value only — never to deposits (deferred) and never line-items operator/driver rates (suppliers price them into their rates in MVP).

Worked examples (agents: these are test vectors):

Scenario	Rates set	d	Winning scheme	hire_value	service_fee	payout
Excavator, daily ₦80k only	d	3	daily ×3	₦240,000	12% = ₦28,800	₦211,200
Excavator, ₦80k/d + ₦450k/w	14	—	weekly ×2 = ₦900k < daily ×14 = ₦1.12M	₦900,000	10% = ₦90,000	₦810,000
Same listing, 8 days	—	8	min(8×80k=640k, 2×450k=900k) = daily ×8	₦640,000	12% = ₦76,800	₦563,200
Generator ₦15k/day	d	1	daily	₦15,000	max(12%×15k=1,800, 2,500) = ₦2,500	₦12,500
Warehouse ₦350k/month	m	30	monthly	₦350,000	6% = ₦21,000	₦329,000

3.2 Payments (D-006 — collect-only)

- On supplier acceptance the hirer receives a Paystack checkout (card / bank transfer / USSD). **Payment window: 4 hours** from acceptance; up to 3 charge attempts within the window.
- Successful charge (webhook-confirmed, never client-redirect-confirmed) → hire becomes **Confirmed**. Window expiry or 3 failed attempts → auto-cancel (`cancelled_by: system`, `reason payment_expired`), dates released, both parties notified.
- A daily reconciliation job compares the Paystack ledger against payment records; any mismatch alerts Ops. Zero-mismatch is a launch criterion.
- **Payouts:** when a hire completes, a payout record (state `due`) enters the Ops payout queue at `payout_amount`. Founder executes bank transfers weekly, recording a reference per payout. States: `pending` → `due` → `paid`, or `frozen` (dispute). A payout is never created for a hire that did not complete — with one exception (D-015): the withheld day on a ≤72h hirer cancellation becomes a supplier payout (`due`) per §7.6.
- **Refunds** per §7.6, issued via Paystack refund API, tracked per hire (`none/pending/completed/failed`); failures alert Ops.

3.3 Money invariants (system-wide)

- Per hire: `collected - refunded - paid_out - retained_fee ≡ 0` at all terminal states.
- Financial fields never mutate after acceptance; corrections happen via Refund records and Ops payout adjustments, all event-logged.
- All monetary display follows design-system money rules (Plex Mono, full value on transactional screens; D-014 audience separation).

4. Users, Roles & Verification

4.1 Roles & account levels

- Role flags: `is_supplier`, `is_hirer`. Registration defaults hirer ON / supplier OFF. Activating supplier

requires completing the Business Profile (business name, description, bank details) before any listing can go Live. Dual accounts switch modes in UI; one identity, one wallet of records.

- **Account levels** (user-facing "Verification"):

Capability	Basic (OTP+email)	Verified (ID approved)	Business Verified (CAC approved)
Browse, message, enquire	✓	✓	✓
Request hires	✓ ≤ ₦250,000 hire value	✓ unlimited	✓ unlimited
Publish listings	✗	✓	✓
Storefront "Business Verified" shield	✗	✗	✓

The ₦250k Basic cap is evaluated at request time against `hire_value`; the request screen surfaces the verification prompt when exceeded.

4.2 Registration & authentication

- Register: full name, email (unique), Nigerian phone (unique, E.164 normalised), password ≥8 chars incl. 1 uppercase + 1 number. **Real SMS OTP via Termii** (6 digits, 10-min expiry, 3 resends/hour, 5 verify attempts then new code required). Welcome email via Resend.
- Login: email+password → JWT access (60 min) + refresh (7 days, rotating, blacklist on logout). 5 failed logins / 15-min IP lockout. Password reset: single-use emailed link, 1-hour expiry, invalidates sessions.
- JWT payload: `user_id, is_supplier, is_hirer, account_level, is_active`.
- One account per person (PB §2.4). Suspension blocks login, hides listings, freezes payouts. Deletion is soft (30-day recovery) then hard, except: financial records retained 7 years, verification documents 5 years (NDPR).

4.3 Verification (real-but-manual, 03 §5)

- Identity: one document (NIN slip / passport / driver's licence / voter's card), JPEG/PNG/PDF ≤5MB → private R2 bucket; pre-signed access only; enters Ops verification queue; approve → Verified, or reject with mandatory reason (user notified, may resubmit). SLA 12h.
- Business: CAC certificate + RC number, same queue → Business Verified.
- **Acceptance checks:** unverified supplier cannot publish; Basic hirer blocked >₦250k with upgrade prompt; rejected user sees the reason and can resubmit; docs never publicly reachable.

5. Supply — Yards, Listings, Storefronts

5.1 Yards

- Fields: name, pin (geographic point), address text, city. Supplier CRUD in portal; a yard with listings cannot be deleted (rename/move only). Listings attach to 0..1 yard and inherit its coordinates; detached listings carry their own pin. Moving a listing between yards updates the map immediately.
- Auto-yard inference: a new listing pinned within 100m of the supplier's existing yard prompts "Add to {yard}?" . Yard creation is offered during supplier onboarding and inline in the listing form.

5.2 Listings

- **Creation (6 steps):** ① class + asset type (from doc 05 launch set; "Other" routes to Ops review) → ② title ≤120 chars, description ≥50 chars, specs per template (required spec fields gate publishing) → ③ pricing: daily ₦ required; weekly/monthly optional; `unit_count` ≥1 (labels per unit optional) → ④ photos 1–10

(JPEG/PNG/WebP ≤10MB each; drag-reorder; cover select) → ⑤ location: yard chip OR pin-drop OR address geocode (pin-drop-first UX, D-013) → ⑥ review & publish.

- **Statuses:** Draft → Live ⇌ Paused → Archived; Removed (Ops only, reason required, supplier notified). Archived/Removed listings preserve hire history; hard delete never permitted once hire records exist. Editing a Live listing never alters locked terms of existing hires.
- **Tiers:** Basic (automatic at publish) / Verified / Inspected — upper tiers manually awarded via Ops in MVP (PB §3.4–3.5 video/inspection machinery activates Phase 3).
- **Spec templates (doc 05 is normative):** per class+type; field kinds number/text/select/multi/boolean with units; required fields gate Draft→Live; one ★ filterable headline spec per class (operating_weight · payload_capacity · floor_area · container_capacity TEU · area). Completeness score stored (ranking input later, not MVP-visible).
- **Reports:** authenticated hirers report a listing (fraudulent / inaccurate / inappropriate / duplicate / unavailable). Reports never auto-hide a listing; 3 reports in 30 days flag priority review; Ops queue shows sibling listings of the same supplier.
- **Acceptance checks:** publish blocked without daily price, ≥1 photo, valid location, required specs; Live→edit keeps existing hires' terms intact; report #3 inside 30 days sets the priority flag.

5.3 Storefronts

- Public, read-only company page per supplier: logo, business name, verification badge, member-since, about, yards as mini-map cards, live listings with class filter chips, **Message CTA** → general enquiry conversation (no listing attached).
- Entry points: listing detail supplier card, Yard Sheet footer, conversation avatar. **No hire CTA on the storefront** — hires always flow through a listing (pricing must be unambiguous).
- Suspended supplier → storefront and listings hidden together.

6. Discovery — Map & Search

- Hirer app home = full-screen map (**MapLibre GL + OpenFreeMap tiles, D-013**), centred on user location; permission denied → Lagos Island (6.4541, 3.3947). **Guest browsing permitted** (map, listings, storefronts); protected actions (enquire, request, report) redirect to auth.
- **Pin hierarchy (normative, Fleet UX Spec):**
 - *Asset Pin* — a solo (yardless or single-listing-yard) Live listing; class-coloured teardrop with class glyph.
 - *Yard Pin* — ≥2 Live listings sharing a yard; supplier logo squircle + count badge + ≤3 class dots + verification tick. **Semantic: never dissolves at any zoom.** Tap → **Yard Sheet**: listings grouped by class, one distance shown, price-from + availability per row, storefront link.
 - *Cluster* — spatial collision of distinct pins at low zoom; neutral circle + count; tap = zoom (max zoom → list of contents). Clusters dissolve on zoom; yards don't.
- Filtering updates yard badges to **matching counts**; zero-match yards dim to 40% opacity (never vanish).
- **Search params:** viewport/bbox + radius (5/10/25/50/100km/custom) around user or searched point; class filter; text (title+description); daily-price min/max; the ★ spec filter per class. Distance is computed server-side (PostGIS geography) and shown as 4.3 km; never client-estimated.
- Results list view toggles **By Asset** (flat, with "+N more at this yard" sub-lines) / **By Location** (yard cards interleaved with solo listings).
- **Acceptance checks:** same supplier+coordinates ⇒ one yard pin at every zoom; two suppliers at one coordinate ⇒ two pins (spiderfy at max zoom), never merged; filtered counts authoritative from server; map search P95 <500ms.

7. The Hire Lifecycle

7.1 Request

- Hirer selects start/end dates on listing detail; dates with no availability are blocked in the picker. Preview shows winning scheme ("14 days → 2 × weekly — best price") and **the total payable only** (D-014). Optional note to supplier. Terms acknowledgment checkbox (ToS + hire terms).
- Availability (unit-aware):** a date range is available iff for the listing, `overlapping_holds < unit_count`, where holding states = **Confirmed, On Hire, and Accepted with unexpired payment window**.
`Requested` does NOT hold dates.
- Multiple overlapping `Requested` hires are allowed — **first-to-pay wins**. When a hire enters `Confirmed`, overlapping `Requested`/`Accepted` hires that now exceed remaining capacity are auto-declined (`no_longer_available`), with notifications.
- Basic-level cap: requests with `hire_value > ₦250,000` are blocked with a verification prompt (§4.1).

7.2 Supplier response

- Supplier sees the request with full financial breakdown, hirer profile (name, account level, completed-hire count), dates, note.
- Accept → terms lock, payment link issued, dates soft-held 4h, hire conversation auto-created. Accepting a listing with operator/driver-included triggers the responsibility acknowledgment (PB §12.2), recorded on the hire.
- Decline → mandatory reason (canned options + free text); dates unaffected (they were never held).
- No response in **24h** → automated **Expired**; hirer notified and nudged toward similar listings.

7.3 Hire state machine (normative)

#	State	Entry	Exits
1	Requested	Hirer submits	→ Accepted (supplier) · → Declined (supplier, reason) · → Expired (24h timer) · → Cancelled (hirer withdraws)
2	Accepted	Supplier accepts; terms locked; payment link issued; dates soft-held	→ Confirmed (payment webhook) · → Cancelled <code>system/</code> <code>payment_expired</code> (4h timer / 3 failed attempts) · → Cancelled (either party, pre-payment, free)
3	Confirmed	Payment received	→ On Hire (start date + on-hire Handover Record, or auto at start+24h) · → Cancelled (refund rules §7.6)
4	On Hire	Start reached / handover confirmed	→ Completed (end passed + off-hire record, or auto at end+48h undisputed) · → In Dispute
5	Completed	Off-hire confirmed	Payout becomes due. Terminal state.
6	Declined / Expired / Cancelled	Per above; <code>cancelled_by</code> ∈ { <code>hirer</code> , <code>supplier</code> , <code>ops</code> , <code>system</code> } + reason	Terminal states; dates released immediately
7	In Dispute	Either party flags during On Hire or ≤72h after end	Payout frozen; founder mediates; Ops resolves → Completed or Cancelled (+refund/payout adjustment, event-logged)

- All timers (24h, 4h, auto-promotions) run on the task queue, idempotent and crash-safe.
- Every transition appends to an immutable hire event log** (timestamp, actor, from→to, metadata) — rendered as the status timeline to both parties and as evidence in disputes. No transition happens outside the state machine; no state is written without its event.

7.4 Handover Records (lite)

- Two per hire: **on-hire** and **off-hire**. Either party initiates, the counterparty confirms in-app. Contents: ≥2 photos, class-specific reading (hour meter — Plant; odometer — Trucks; none — spaces, which use condition/occupancy notes; per 03 §7), optional notes, both parties' confirmation taps.

- Non-blocking in MVP (no deadlines or penalties), but refusal to confirm weakens that party's dispute position (PB §10.4 defaults applied as Ops guidance).
- Photos go to the private bucket; visible only to the two parties and Ops.

7.5 Extensions

Not in MVP — cancel-and-rebook. `parent_hire` reserved for Phase 2.

7.6 Cancellation & refunds (normative)

Scenario	Outcome
Pre-payment (Requested/ Accepted), either party	Free; no money moved
Hirer cancels post-payment, >72h before start	100% refund
Hirer cancels post-payment, ≤72h before start	Refund = <code>hire_value</code> – one day's daily-equivalent – ~1.5% processing. The withheld day becomes a supplier payout (<code>due</code>)
Supplier cancels post-payment (any time)	Hirer refunded 100% (Terminal absorbs processing cost); supplier strike (3 strikes → suspension review). No cash penalties in MVP
Hirer no-show (handover +24h)	Treated as hirer ≤72h cancellation
Supplier no-show (asset unavailable at handover)	Treated as supplier cancellation

Handover Record absence is the no-show evidence. "One day's daily-equivalent" = `hire_value / duration_days`, rounded to the kobo.

8. Messaging

- **Conversations:** Enquiry (listing-level, or storefront-level "general") and Hire conversations (auto-created at Accepted). Parties: the supplier and hirer only; Ops may view in dispute context. One enquiry conversation per (listing, hirer); one conversation per hire.
- Delivery: message of record persists via REST; **realtime fan-out via Ably (D-011)**; graceful degradation to 15-second polling. History permanent. Unread counts per conversation + aggregate badge.
- **Contact masking until the related hire is Confirmed:** in enquiry and pre-payment hire conversations, Nigerian phone patterns and email addresses are redacted server-side at write time (rendered as 0803... 0 + "Unlocks after payment" explainer — informative styling, never punitive). Post-payment hire conversations are unmasked. No keyword surveillance beyond this regex (deliberate privacy posture). Bypass-solicitation reports → Ops ladder: warn / 7-day suspension / ban (PB §11.5).
- Conversation list rows: counterparty + verification badge, listing thumbnail + title (or "General enquiry"), yard name, last message preview, timestamp, unread badge.
- **Acceptance checks:** third parties get 404 on others' conversations; masking applies pre-payment and lifts post-payment for that hire's conversation only; enquiry conversations stay masked indefinitely; Ably outage degrades to polling without message loss.

9. Notifications (MVP = in-app badge + email; push deferred Phase 3)

Event	Hirer	Supplier	Channel
Hire requested	—	✓	badge + email
Accepted (pay now, 4h)	✓	—	badge + email (deadline prominent)

Declined / Expired	✓	— / reminder at 20h	badge + email
Payment confirmed	✓ receipt (no fee shown)	✓ (with payout figure)	badge + email
Payment window expiring (60 min left)	✓	—	badge + email
Auto-cancelled (payment expired)	✓	✓	badge + email
Cancellation (any)	✓	✓	badge + email
Handover awaiting confirmation	counterparty	counterparty	badge
Hire completed	✓	✓ (+payout queued)	badge + email
Payout paid	—	✓ (reference)	email
New message	✓	✓	badge (realtime)
Verification approved/rejected	✓	✓	badge + email
Supplier strike / listing removed	—	✓ (reason)	email

Supplier notification preferences (per-type toggles, default ON) apply to email only; badges always accrue.

10. Surfaces

10.1 Hirer Mobile App ("Terminal app") — screens normative in ux/02

Map/Home (pin hierarchy, filter bar, search) · Yard Sheet · Listing Detail (gallery, dynamic SpecTable, pricing grid + best-price hint, supplier card → Storefront, availability calendar, Enquire / Request to Hire, report via overflow) · Request flow (dates → preview [total only, D-014] → note → acknowledge → confirm) · Pay (Paystack in in-app browser + CountdownPill; status resolved by webhook polling, never the redirect) · My Hires (tabs Requested / Upcoming / On Hire / History) · Hire Detail (EventTimeline, LockedTerms hirer variant, handover records, conversation link, cancel) · Handover capture (camera-first, reading field, confirm) · Messages + Conversation (MaskedContact notices) · Storefront · Profile (verification status/upload, role activation, settings) · Onboarding (role intent, OTP, optional verification prompt). Design language per doc 08 ("Heavy Duty").

10.2 Supplier Portal — screens normative in ux/03

/dashboard (pending requests with 24h countdowns, awaiting-payment, unread, month earnings, next payout, activity feed) · /assets (DataTable: status/class/yard filters, bulk pause/archive/assign-yard, **Duplicate Listing**) · /assets/new & /assets/:id (6-step Stepper; spec template renders per class) · /yards (CRUD) · /hires (DataTable + filters) · /hires/calendar (CalendarGantt: yard-grouped rows, unit-utilisation bars, pending dashed) · /hires/:id (LockedTerms supplier variant, accept/decline+reason, handover records, chat panel) · /messages · /storefront (public preview + business profile edit) · /settings (personal, business, bank masked, notification prefs, password, verification). Acknowledgment checkboxes where required (operator/driver hires).

10.3 Ops Console (Django Admin, staff-only + 2FA)

Dashboard (GMV, fees collected, payout liability, hires by state, new users, reconciliation status) · Verification queue (doc viewer, approve/reject+reason) · Payout queue (due payouts, bank details, mark-paid+reference, freeze/unfreeze) · Listings admin (tier award, pause/remove+reason, reports queue with sibling context) · Hires admin (event log, admin cancel, dispute resolution: → Completed or → Cancelled + refund/payout adjustments) · Users (suspend/reactivate, strikes, notes) · Reconciliation report. **Every Ops action lands in the event log / Django LogEntry.**

11. End-to-End Journeys (acceptance narratives; expanded in ux/00)

1. **Fleet supplier onboards** — Adaeze (BNN, 25 assets, 2 yards): register → OTP → supplier activation + business profile → ID verified by Ops ≤12h → creates Apapa + PH yards → first listing via 6-step form (yard chip, specs template) → duplicates for the fleet, unit_count for the 8 identical tippers → map shows two BNN yard pins, badges 20 and 5. *Under an hour of work.*
2. **Hirer discovers and pays** — Emeka (site manager): installs app → map → filters Plant → yard pin 12.4km → Yard Sheet → CAT 320 listing → 14 days → "2 × weekly" preview ₦900,000 total → registers + OTP mid-flow → requests → Tunde accepts in 3h → Emeka pays via transfer within window → Confirmed; conversation unmask; receipt (no fee shown).
3. **The hire runs** — on-hire handover (photos + hour meter, both confirm) → On Hire → off-hire record at return → Completed → payout due ₦810,000 → founder pays Friday batch, records reference → supplier emailed.
4. **Things go wrong (each a test scenario)**: supplier silent 24h → Expired + hirer nudge · payment window lapses → auto-cancel + dates release · hirer cancels 5 days out → 100% refund · supplier cancels paid hire → full refund + strike · damage claim → In Dispute → payout frozen → Ops resolves with adjustment · fake listing reported 3× → priority review → Removed + supplier note.
5. **Ops weekly rhythm** — verification queue twice daily (12h SLA), payout batch Fridays, reports queue, reconciliation green daily, weekly metrics digest email.

12. Non-Functional Requirements

- **Performance**: map search <500ms P95 @ 500 in-radius listings; non-geo API <200ms P95; webhook→state <60s; realtime message <300ms P95; photo upload <5s/5MB on 4G; app cold start to interactive map <4s mid-tier Android.
- **Security**: HTTPS+HSTS; JWT 1h/7d rotating+blacklist; bcrypt cost ≥12; bank numbers encrypted at rest (AES-256/Fernet); private R2 + short-lived presigned GETs for docs/handover photos; Paystack HMAC-SHA512 webhook verification + event dedup; auth rate limits (§4.2); CORS allowlist; Ops Console staff-only + 2FA; no secrets in clients.
- **Money correctness**: §3.3 invariants; daily reconciliation; immutable financials; append-only event log.
- **Reliability**: 99.5% uptime; Sentry on all four surfaces; daily DB backups, 7-day retention; Ably→polling degradation; timers queue-backed and idempotent (a missed sweep run self-heals next run).
- **Compliance (NDPR)**: ToS/Privacy at registration; per-hire acknowledgments; retention — financial 7y, verification docs 5y; soft delete 30d → hard delete with carve-outs; data stored in NDPR-compatible jurisdictions.

13. Launch Acceptance (gate, from 03 §8)

- **Supply**: ≥40 Live listings (≥25 construction corridor, ≥15 port corridor), ≥8 storefronts with 3+ listings.
- **Within 60 days**: ≥25 paid hires; ≥₦500k service fees collected; ≥30% hirer repeat rate.
- **Product**: supplier register→Live <1 day; hirer install→request <10 min; full loop incl. handover unaided.
- **Technical**: NFRs green; zero unreconciled payment states; the §11.4 failure scenarios all pass as tests.

— End of FSD v2.1 —